DERIVATIVES QUIZ

BLAKE FARMAN

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Name: Solutions

1. Use the **limit definition**

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

to compute the derivative of

$$f(x) = \frac{1}{x}.$$

Check your answer using the Derivative Rules.

$$f'(x) = \lim_{h \to 0} \frac{\frac{1}{(x+h)} - \frac{1}{x}}{h} = \lim_{h \to 0} \frac{1}{h}\left(\frac{x}{x(x+h)} - \frac{x+h}{(x+h)x}\right)$$

$$= \lim_{h \to 0} \frac{x - x - h}{hx(x+h)}$$

$$= \lim_{h \to 0} \frac{-h}{hx(x+h)}$$

$$= \lim_{h \to 0} \frac{-1}{x(x+h)}$$

$$= \frac{-1}{x(x+0)}$$

$$= \boxed{\frac{-1}{x^2}}$$

Check: $\frac{d}{dx}\frac{1}{x} = \frac{\frac{d}{dx}(1)x - 1\frac{d}{dx}(x)}{x^2} = \frac{0-1}{x^2} = \frac{-1}{x^2}$

or

$$\frac{d}{dx}\frac{1}{x} = \frac{d}{dx}x^{-1} = -x^{-2} = \frac{-1}{x^2}.$$